

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:

Measure network performance using key metrics with simulation tools
(BL4,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks:30

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

Question 1: Connectivity, Devices & Security for a Remote Rural Branch Office

1. Understanding of the Problem & Constraints

- No wired/fibre broadband is available at the site, so the primary constraint is last-mile connectivity, not LAN design.
- Workload is bandwidth- and latency-sensitive: cloud applications (SaaS, ERP), real-time video conferencing, and secure links to headquarters (HQ).
- Business requires uninterrupted operations, implying a need for redundancy/failover, not just raw speed.
- Budget is limited, which rules out expensive options such as leased lines or fibre trenching, and favours OPEX-based, subscription models.
- Solution must be future-scalable as the branch grows (more staff, more traffic, possibly more branches).

2. Problem Decomposition

- **Sub-problem 1:** Last-mile internet access for a rural site with no wired infrastructure.
- **Sub-problem 2:** Local networking devices (routing, switching, Wi-Fi) to distribute connectivity within the office.
- **Sub-problem 3:** Redundancy/failover to guarantee uninterrupted operation if the primary link degrades.
- **Sub-problem 4:** Security mechanisms for safe communication with HQ and protection of company data.
- **Sub-problem 5:** Cost control while keeping the design scalable for future growth.

3. Proposed Solution (Application of Constraints)

Recommended architecture: a Low-Earth-Orbit (LEO) satellite link (e.g., Starlink Business) as the primary WAN connection, backed by a 4G/5G LTE router as a secondary failover link, feeding an SD-WAN-capable router that load-balances/fails over automatically, a managed switch, a Wi-Fi access point, and a UTM firewall with a site-to-site VPN to HQ.

Internet connectivity

- Primary: LEO satellite broadband — does not depend on local wired infrastructure, delivers tens to a few hundred Mbps download and sufficient upload for HD video conferencing, and has far lower latency than traditional geostationary satellite.
- Secondary/backup: 4G/5G cellular router — inexpensive, quick to deploy, and automatically takes over if the satellite link drops, directly addressing the 'uninterrupted operations' requirement.

Networking devices

- Dual-WAN / SD-WAN router to combine the two links, apply automatic failover, and prioritise (QoS) latency-sensitive video-conferencing and VPN traffic over bulk downloads.
- A single managed Layer-2/3 switch and one or two enterprise-grade Wi-Fi access points are enough for a small branch, keeping capital cost low while allowing more APs to be added as headcount grows.

Security mechanisms

- Site-to-site IPsec VPN tunnel between the branch router and HQ to encrypt all inter-office traffic.
- UTM/next-generation firewall at the branch edge for stateful inspection, intrusion prevention, and web/content filtering.
- WPA3-Enterprise or WPA2-Enterprise Wi-Fi with 802.1X authentication so only authorised staff devices join the network.
- Multi-factor authentication (MFA) for cloud application and VPN logins to protect access even if the link is intercepted.

4. Logical Reasoning & Strategy

- Bandwidth: LEO satellite typically provides enough throughput for video conferencing (which needs only a few Mbps per stream) and cloud SaaS use, so it satisfies the workload without needing a costly leased line.
- Reliability: dual-WAN failover directly targets the 'uninterrupted business operations' constraint — if satellite weather-fade or an outage occurs, LTE keeps the office online.
- Cost: satellite and LTE are both subscription (OPEX) services with no trenching or civil work, which fits a limited budget far better than laying fibre to a remote site.
- Scalability: SD-WAN and cloud-managed Wi-Fi/switches can be scaled by adding APs, bandwidth tiers, or additional branch sites without re-architecting the network.

5. Solution Accuracy, Feasibility & Constraint Trade-offs

- The design is feasible with commercially available, off-the-shelf equipment and services, so it can be deployed quickly at a rural site.
- Trade-off acknowledged: satellite links can have higher latency and occasional weather-related degradation compared to fibre; the LTE failover and QoS policies mitigate this rather than eliminate it entirely.
- Trade-off acknowledged: as headcount grows significantly, the branch may eventually need a dedicated leased line, but the proposed design defers that capital cost until it is actually justified.

6. Creativity & Innovation

- Using SD-WAN to actively load-balance across satellite and cellular links (rather than pure cold-standby failover) so both circuits contribute to throughput during normal operation.
- Applying application-aware QoS to guarantee bandwidth for video conferencing and VPN traffic even during network congestion, an innovative way to meet performance needs without buying more bandwidth.

Question 2: Campus-Wide Wireless Network Design for a Large University

1. Understanding of the Problem & Constraints

- Very large number of concurrent users (3,000+ students) across diverse zones: classrooms, laboratories, libraries, hostels.
- Signal interference must be minimised despite high user density and many overlapping coverage areas.
- Secure access is mandatory — the network must authenticate and authorise legitimate users.
- Budget allows only a limited number of access points (APs), so AP count and placement must be optimised, not maximised.

2. Problem Decomposition

- **Sub-problem 1:** Capacity planning — estimating concurrent client load per zone to size the number of APs needed.
- **Sub-problem 2:** AP placement and coverage design for classrooms, labs, libraries and hostels.
- **Sub-problem 3:** Frequency band and channel planning to minimise co-channel and adjacent-channel interference.
- **Sub-problem 4:** Authentication and security architecture for students, faculty and guests.
- **Sub-problem 5:** Cost optimisation while still meeting coverage and performance targets.

3. Proposed Solution (Application of Constraints)

Recommended design: Wi-Fi 6 (802.11ax) access points deployed at higher density in high-traffic zones and lower density in low-traffic zones, using mostly the 5 GHz band for capacity with 2.4 GHz reserved for range/legacy devices, centrally managed by a wireless LAN controller (or cloud controller) for band-steering and load balancing, secured with WPA3-Enterprise/802.1X.

Access point placement

- Zone-based density: more APs (ceiling-mounted, higher client-capacity) in libraries, large lecture halls and computer labs where user density peaks; fewer APs with wider coverage in hostels and corridors.
- Site-survey-driven placement to avoid dead zones and to overlap coverage cells by roughly 15–20% for seamless roaming, without excessive overlap that wastes AP budget.
- Directional/patch antennas in long corridors and auditoriums; omnidirectional APs in open study areas.

Frequency band and channel selection

- Primary client capacity on the 5 GHz band, which has many more non-overlapping 20/40 MHz channels than 2.4 GHz, reducing co-channel interference in dense deployments.
- 2.4 GHz kept active mainly for range and for older IoT/legacy devices, with lower transmit power to limit its interference footprint.
- Wi-Fi 6 features (OFDMA, MU-MIMO, BSS colouring) used to let each AP serve more simultaneous clients efficiently, directly reducing the number of APs required for the same user load.
- Automatic channel/power management (via the WLAN controller) to adapt to interference in real time instead of static manual planning.

Authentication and security

- WPA3-Enterprise with 802.1X and a RADIUS server tied to the university directory (single sign-on) for students and staff — similar to the eduroam model.
- Separate SSIDs/VLANs for students, faculty, and guests, with the guest network isolated from academic/administrative resources.
- Captive portal with time-limited access for visitors, keeping the secure enterprise SSID reserved for verified university accounts.

4. Logical Reasoning & Strategy

- Coverage vs. AP count: prioritising AP density where usage is highest (library, labs) and accepting lighter coverage in low-usage areas keeps the total AP count within budget while still meeting peak demand where it matters most.
- Performance: Wi-Fi 6's OFDMA/MU-MIMO lets a single AP efficiently serve dozens of simultaneous low-bandwidth client sessions (typical of students browsing/streaming), reducing the number of APs needed compared to older standards.
- Interference minimisation: concentrating capacity in 5 GHz with a proper channel plan and centralised RF management avoids the channel congestion that would occur if everything ran on the crowded 2.4 GHz band.
- Security: enterprise-grade 802.1X authentication is chosen over simple pre-shared keys because a shared password cannot be revoked per-user at this scale, and it supports auditing/accountability.

5. Solution Accuracy, Feasibility & Constraint Trade-offs

- The design is feasible using commercial enterprise Wi-Fi 6 hardware and a standard AAA/RADIUS back end that most universities already operate for their directory service.
- Trade-off acknowledged: hostels and open outdoor areas will have comparatively lower AP density, so peak-hour throughput there will be lower than in academic buildings — an acceptable trade-off given usage patterns.
- The centralised controller approach adds a modest licensing/hardware cost but pays for itself by reducing the number of physical APs needed through smarter RF management.

6. Creativity & Innovation

- Using AI/analytics-driven RF management (available in most modern cloud WLAN controllers) to auto-tune channels and power in real time as classes change and student density shifts through the day.
- Phased/adaptive deployment: start with APs in the highest-traffic zones, then use built-in analytics (client count, airtime utilisation) to identify where additional APs give the best return before spending the remaining budget.

Question 3: Secure Network Architecture for a Financial Institution

1. Understanding of the Problem & Constraints

- The institution must strengthen cybersecurity while employees still need access to sensitive customer data and online banking systems.
- Strict regulatory/security compliance must be met (e.g., data protection and banking security standards).
- The redesign must not significantly degrade network performance for day-to-day banking operations.
- Operational costs must be controlled — the solution cannot rely purely on the most expensive enterprise-grade tools everywhere.

2. Problem Decomposition

- **Sub-problem 1:** Network segmentation to separate systems of different sensitivity levels.
- **Sub-problem 2:** Firewall deployment strategy (perimeter and internal) to control traffic between segments.
- **Sub-problem 3:** Authentication and access-control methods for employees accessing sensitive systems.
- **Sub-problem 4:** Intrusion detection/prevention and monitoring to catch and respond to threats.
- **Sub-problem 5:** Balancing all of the above against performance and cost constraints.

3. Proposed Solution (Application of Constraints)

Recommended architecture: a zoned/segmented network (DMZ, internal corporate zone, restricted core-banking zone, guest zone) separated by next-generation firewalls, protected by role-based multi-factor authentication, and monitored by an IDS/IPS feeding a SIEM for compliance logging.

Network segmentation

- DMZ zone hosts internet-facing services (online banking web/app servers), isolated from internal systems.
- Internal corporate zone for general employee workstations and email/office applications.
- Restricted core-banking zone (customer databases, transaction-processing servers) isolated with its own VLAN and stricter access rules — only specific applications/roles can reach it.
- Separate guest/vendor VLAN with no route to any internal or restricted zone.
- Micro-segmentation around the most critical assets (customer data stores) so that even a compromised internal workstation cannot directly reach them.

Firewall deployment

- Perimeter next-generation firewall (NGFW) between the internet and the DMZ, performing stateful inspection, deep packet inspection and application-layer filtering of banking traffic.
- Internal segmentation firewalls between the corporate zone and the restricted core-banking zone, so lateral movement from a compromised employee device is blocked by policy, not just detected after the fact.
- Hardware-accelerated firewall appliances used at these choke points specifically to avoid the performance penalty of software-only inspection on high transaction volumes.

Authentication and access control

- Multi-factor authentication (MFA) for all employees accessing customer data or the core-banking zone.
- Role-based access control (RBAC) so employees only see the data required for their job function (least privilege).
- Privileged access management (PAM) with session recording for administrator accounts on core-banking servers.
- Certificate-based authentication for server-to-server communication inside the restricted zone.

Intrusion detection and monitoring

- Network-based IDS/IPS at the DMZ boundary and at the entrance to the restricted core-banking zone to detect and block known attack patterns in real time.
- Centralised SIEM aggregating firewall, IDS/IPS, and authentication logs — both for threat correlation and for the audit trail required by regulatory compliance.
- Regular vulnerability scanning and penetration testing scheduled to avoid peak transaction hours, so it does not affect production performance.

4. Logical Reasoning & Strategy

- Security vs. performance: hardware-accelerated NGFWs and dedicated IPS appliances are placed only at key segmentation boundaries (not on every internal link), so inspection overhead is limited to where it matters most, protecting throughput for routine transactions.
- Compliance: segmentation, MFA, encryption and centralised logging directly map to common financial-sector compliance requirements (access control, audit trails, data protection), satisfying the 'strict security policies' constraint.
- Cost control: segmentation and internal firewalls reuse VLAN capability on existing switches wherever possible, and open-source/lower-cost SIEM options can be used for log aggregation, reserving premium NGFW spend for the highest-risk boundaries (perimeter and core-banking zone).
- Least privilege (RBAC + micro-segmentation) reduces the attack surface without adding new hardware, addressing security at effectively zero extra operational cost.

5. Solution Accuracy, Feasibility & Constraint Trade-offs

- The design is feasible with standard enterprise networking and security equipment already common in the financial sector, so it can be implemented incrementally without a full network rebuild.
- Trade-off acknowledged: internal segmentation firewalls add a small amount of latency between zones; this is mitigated by using hardware-accelerated appliances and by segmenting only where sensitivity genuinely differs, not at every hop.
- Trade-off acknowledged: strong MFA/RBAC adds a login-time step for employees; this is an acceptable and typically mandated cost given the sensitivity of customer financial data.

6. Creativity & Innovation

- Using micro-segmentation (rather than only coarse VLAN zoning) around the customer database specifically, so that even a fully compromised employee laptop in the corporate zone cannot directly reach it — an extra layer beyond the minimum ask.
- Scheduling vulnerability scans and heavier security jobs outside peak banking hours, an operational trick that reconciles the 'no significant performance impact' constraint with thorough security testing.